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4th Industrial Revolution Design Through Lean Foundation

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Abstract

Industry 4.0 was introduced with the hope of transitioning to the next industrial revolution. I4.0 is a vision of increasing flexibility and efficiency within a system of low complexity. Despite this intention, Industry 4.0 has required users to understand complex information and communication systems, limiting usability and therefore not necessarily satisfying the original vision. However, application of principles from the Lean Production Systems originally developed in the 1930's in combination with Industry 4.0, may help to reach its desired goals. Through a literature review, this paper discusses the current state of art of Industry 4.0 and some Lean Production Systems principles. The paper focuses on a model combining Lean Production Systems with Industry 4.0, that was created in response to the shortcomings of the anticipated industrial revolution. With a Lean Production System as a foundation, the six design principles of I4.0 of interoperability, virtualization, decentralization, real-time capability, service orientation, and modularity can be better implemented to reach the vision of highly flexible and low complexity systems. In return, horizontal and vertical integration within Industry 4.0 can support and accelerate Lean principles by reducing human error and increasing intercommunications between the systems, resulting in a productive integration. The specific principles of Lean that could boost Industry 4.0 will be discussed, demonstrating why a company should be Lean, meaning value adding processes and stakeholder orientation are at its core.

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1. Introduction

Research has shown that Lean production systems (LPS) and Industry 4.0 (I4.0) are influential factors on one another [1]. This paper will detail the main principles of Lean that should be present in the foundation of an enterprise for better implementation of I4.0 [1]. I4.0 and LPS create the opportunity to increase stakeholder value, allowing enterprises to stay competitive in a global market. The increase of man to machine interaction suggested by I4.0 requires an enterprise to address socio-technical issues and requires the company values to be at the forefront of all human or machine-made decisions [2]. I4.0 and LPS have overlaps in targets with regards to efficiency and productivity, however the different leveraging tools to reach these goals are concentrated on IT and human centricity respectively. The model proposed in this paper was designed based on the shortcomings identified through literature review on how to better implement I4.0 with LPS principles.

2. State of Art

2.1. Lean Production Systems (LPS)

Lean production systems are an enterprise specific set of rules to improve processes, by aligning the employees with a common vision [2]. Implementing Lean principles that are appropriate for an organization, such as avoiding waste and orienting toward the customer, will achieve a higher performance that increases the competitiveness of the enterprise [2]. There are various principles in a LPS, standardization, zero defect, pull and flow production, and employee orientation are the principles that stood out in the literature review in section 3 as shortcomings faced during the

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implementation of I4.0. The other LPS principles of continuous improvement, visual management, and avoidance of waste are not explicitly excluded but are not further discussed in the paper. The following sections describe the current state of art of these four principles mentioned above.

2.1.1. Standardization

Standardization is a principle designed to create a common technical set of rules agreed upon by multiple parties including producers, users, interest groups, and governmental agencies. To generate a common understanding of a product, process, or service and to decrease deviations and undesired actions, the rules are to be followed by all [2]. 5S is a common method employed by Lean for achieving order and cleanliness by standardization [3]

2.1.2. Zero defect

Zero defect principle is the avoidance of defects or their transmission to subsequent downstream processes [2]. If not corrected or stopped at the point of origin, these defects are easily replicated, resulting in rework and an exponential rise in cost. [4]. Autonomation one of zero defect methods is the ability of a machine to operate without human surveillance [2]. This gives equipment the ability to distinguish good parts from bad autonomously, without being monitored by an operator and to stop production if defects are encountered [5].

2.1.3. Pull and flow production

Pull and flow production principle creates a continuous flow of material and information across the entire value chain, orienting towards the demands of the customer. Its aim is to achieve short throughput times and a low inventory level without reducing flexibility [2]. One method from pull and flow production is value stream mapping (VSM), this is a powerful visual method used to identify non-value-added activity. It helps to understand the current state of product flow and reduce waste to achieve shorter throughput times and increase flexibility. A VSM consists of the entire set of activities from raw material to finished product for a specific product or product family [6].

2.1.4. Employee orientation

Employee orientation is a cultural promotion to reduce waste and decrease defects [2]. The employees understand the processes best and are considered to be the most important resource toward continuous improvement of the organization [2]. To achieve an employee-oriented transformation, there needs to be a change in the behavior and mindset of the organization where everyone works towards a common vision by accepting continuous improvement as routine work. The cultural change is fundamental to an LPS transformed organization.

2.2. Industry 4.0

Industry 4.0 aspires to a high level of connectivity of all processes and products over the entire lifecycle [7]. Within the context of production - man, machine, product, and processes are all self-organizing and connected. This allows an integrated response to changes and incorporates reviving planning into management systems [8]. For business enterprises, this is viewed as a revolution in manufacturing to increase productivity and increase stakeholder value. While the promises of I4.0 have made many great claims, one misconception is that I4.0 is a solution to problems within itself [9]. I4.0 requires a paradigm shift within companies to be innovative, stay competitive, and increase productivity. The current pathway to reaching the high level of autonomy required for this revolution is still not totally clear, despite the proposed Reference Architecture Model for Industry (RAMI 4.0) and the Intelligent Manufacturing Systems Architecture (IMSA). The current lack of understanding and acceptance of the standards in RAMI4.0 is evident in the low number of paper citations [10]. RAMI 4.0 is a map of horizontal and vertical integration across the product life cycle and ensures secured network data [11]. The horizontal and vertical integration of I4.0 increases visibility and communication throughout an entire value chain and therefore suggests new business models based on adding value for all stakeholders [12]. The implementation of I4.0 comes at a financial risk for companies and the investment cost in high tech equipment and IT infrastructure has been a deterrent for many, especially small and medium sized enterprises [13].

Interoperability, virtualization, decentralization, real-time capability, service orientation, and modularity are the six design principles of I4.0 identified from a literature review. Their current state of art will be discussed to better understand the requirements and their missing elements [14].

2.2.1. Interoperability

Interoperability within a production system is the ability to work with connected products and systems without increased effort from the customer [15]. Within the interoperability context, machines can exchange and interpret data among themselves. This level of connectivity creates a new environment and level of integration between services, products and processes that serves as a basis for I4.0 development [16]. There are already copious amounts of data being created by humans, machines, and products. However, due to a lack of standardization within IT protocols, many of these systems are unable to communicate with one another [17] The technological barriers created between segments in a value chain limits visibility between suppliers, producers, and customers.

2.2.2. Virtualization

Virtualization is the graphic representation of the physical world within a production environment that allows for the visualization of production planning and control (PPC) of products and processes. The creation of a virtual or “digital

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twin" of not just products and processes, but of an entire value chain, allows for an end-to-end view of a product life cycle. This digital twin helps to reduce the time required for planning of new products by allowing designers to see possible errors in setup or simulation [18]. The decentralized data acquisition of a digital twin is facing difficulties due to the required standardization of information systems [19]. The lack of standards in data management strain the integration of processes [17,20].

2.2.3. Decentralization

Decentralization in I4.0 is the delegation of decision making to lower hierarchies in the production process, allowing a system to self-organize and react to changes in order to maximize productivity. The purpose of decentralized decision making is to increase the flow of information and to avoid overloading of sophisticated execution blocks with meaningless data. Closed loop systems make decisions autonomously with the ultimate goal of the organization in the foreground [18]. While a rise in microprocessors in manufacturing suggests this decentralization is underway, the inherent problems of decentralization of decisions is complex and presents design challenges for humans and algorithms (i.e. game theory) [21].

2.2.4. Real-time capability

Data collection and real-time processing by the continuous monitoring of product or process is used to reduce reaction time and increase productivity. Real-time capability is a requirement for both decentralized decision making and virtualization. For creating a virtualized CPS, the crossover between a digital and physical world is dependent on accurate physical positioning [22]. There is significant idle time spent on data transfer. For example, in mechanical and engineering plants only 4 percent of throughput time is associated with production, while most of the time is spent on ordering, procurement, delivery etc. [23]. Improvements in these processes will reduce throughput as a result.

2.2.5. Service Orientation

Service orientation refers to granting machines, people, and software access to information and allowing them to make real-time changes in the system based on customer preference. Adding service orientation to business models can create transparency in the value chain and therefore increase the customer focus [22]. This results in a better relationship between B2B, B2C, and allows for efficient knowledge flow across supply chains in the horizontal integration. This shift in business models suggests a transition to an outcome economy, where customers pay for results only.

2.2.6. Modularity

Modularity refers to the standardization across product vendors and communication networks to create systems that work well with one another and require little to no adjustments. The term modularity applies not only to modular products that allow for mass customization after the customer decoupling point, but also to a modular production system that is interchangeable and fast reacting. [22, 24].

The combination of the six design principles are what is required for reaching the vision of I4.0. From the literature review, the state of the art of I4.0 design principles identified their focus and current shortcomings. Changes in the implementation strategy of I4.0 design principles are required for achieving its vision on high flexibility with low complexity.

3. Lean Industry 4.0

Both Lean and I4.0 emphasize process improvement and increased efficiency, but do not consider one another in their implementation. However, one is not mutually exclusive from the other. Table 1 shows the shortcomings of I4.0 that were identified through a literature review and the corresponding Lean principles that could be implemented to overcome these challenges. Not all challenges faced by I4.0 could be identified as having an LPS enabler, data security being one example.

Table one shows the specific shortcomings identified in the literature review that could be overcome with LPS due to their overlap of goals are:

- Inadequate standardization
- Focus on yield instead of reliability
- Integration gap between different departments
- Lack of skilled workforce

A common standard for product, process or services can be established by following standardization principles to avoid undesired activities or deviations. The zero defect principle is the avoidance of defects; this quality drive focuses companies on reliability and the reduction of rework. The integration gap in departments could be overcome by use of the VSM method, to better understand the necessary assets required by stakeholders. Understanding the requirements for value chain segments helps to establish vertical and horizontal integration in the company. Lastly, the culture change achieved through employee orientation under Lean, helps to promote and train the employees, knowing that they are the most important resources for continuous improvement in an organization.

Both Lean and I4.0 strategies aim for many of the same targets, but with different approaches and implementation strategies [34]. An LPS should be established before the initiation of a widespread I4.0 initiative to ensure a stable process as well as a refined value stream [34, 15].

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Table 1: Summary of literature review of I4.0 shortcomings and corresponding Lean principles

Shortcomings of I4.0	Lean Principles
Lack of data security and uniform standards, lack of user transparency, insufficient qualified work force [25]	S, P&F, EO
Inadequate common standards, lack of managing complex products and manufacturing systems, less focused personal and professional development [10]	S, P&F, EO
Integration gap between shop floor and ERP, lack of standard approach in data management, shortage of skilled staff, concern of security, shortcoming in implementation of machine repairs [17]	S, P&F, EO, ZD
Lack of fully coupled human machine interaction, inadequate utilization of available information [26]	P&F, EO
Insufficient standardized processes for automation, needs more focus in education and communication, asynchronous improvement in all departments [20]	S, P&F, EO
Less emphasis on training and contribution from all levels of employee, insufficient established standards, lack of policy supporting technological innovation [27]	S, EO
Inadequate social awareness and acceptance, legal issues, lack of knowledge and trust management [28]	S, EO
Lack of focus on social sustainability (operator 4.0) [29]	EO
Lack of necessary talent, difficulty in coordinating actions, concerns about cybersecurity, lack of a clear business case, challenges with integrating data [30]	S, P&F, EO
Not being led by quality, issue of industrial security risk (unauthorize remote access), lack of a clear understanding of the significance of technology and application [31]	P&F, EO, ZD
Trying to improve yield and not reliability and accuracy, improvement is focused on individual machine as isolated island [32, 33]	P&F, ZD

Standardization (S), Pull and Flow Production (P&F), Employee Orientation (EO), Zero Defect (ZD)

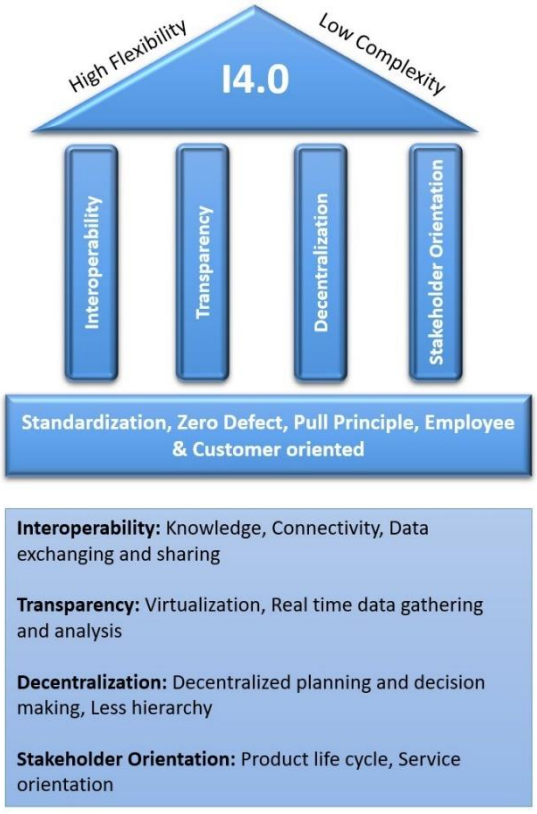


Figure 1: Model for better integration of I4.0 with a foundation of Lean principles.

The design model in figure 1 shows the results of the literature review combined with the coupled design principles of I4.0, and demonstrates that I4.0 can be better implemented with a Lean foundation. If enterprises have a matured LPS consisting of the mentioned principles, the shortcomings of I4.0 can be addressed. The four pillars illustrated in figure 1 are based on the foundation of different Lean principles for the better implementation of the I4.0 vision of high flexibility and low complexity with minimized efforts [35].

Overlaps in the requirements in the six I4.0 design principles allowed us to combine them into the four pillars consisting of: interoperability, transparency, decentralization, and stakeholder orientation. Real time data was combined with decentralization and virtualization because it is a prerequisite for the design principles, as described in section 2.2.4. Virtualization was combined with transparency because of the shared target of displaying and controlling activities within the value chain. The last design principle of modularity is implicitly achieved by implementation of the entire design model in figure 1.

By establishing a Lean foundation within an enterprise, the acceptance of a common vision and set of values by all employees is implied, allowing for easier implementation of change [2]. With Lean principles as a base and the 4 pillars of interoperability, transparency, decentralization and stakeholder orientation, the vision and transformation of high flexibility within a low complex system can be achieved.

3.1. Lean Interoperability

On a semantics level, the meaning of the information transferred must be commonly understood between interfaces. According to the definition from IEEE, interoperability is achieved through standardization [36]. Standardization of semantic protocols between companies has the potential for savings by eliminating waste in data handling. For example, by avoiding the misinterpretation of data by users [37]. An optimized production process achieved through pull and flow production principle focuses on value adding activities, data in the value chain should effortlessly flow through connected systems which could enable interoperability in I4.0. The International Data Space Association, which is a collaboration between Fraunhofer and industry leaders, aims to increase standardization and sovereignty of data spaces for effective interoperable systems [11].

On a vertical level, interoperable systems react to upstream changes in processes and share and confirm data with all relevant stakeholders. This means that in the supply chain,

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customers and suppliers come up with collaborative solutions to the current situation. If logistical issues occur, a coping measure should be enacted throughout the supply chain to adhere to on-time delivery.

A successful implementation of standardization and pull and flow production in the LPS could improve interoperability by eliminating non-value adding activity.

3.2. Lean Transparency

Transparency is a crucial prerequisite for developing flexible production systems in I4.0. A clear view of systems through I4.0 tools allows stakeholders to keep track of product or service development [38]. Value stream mapping (VSM) is a method that aims to create workflow transparency by integrating stakeholders. This makes it easy for stakeholders to get an overview and status updates of a production system at one glance, the virtualization of this process is supported by the digital twin [2].

Real-time data transfer and monitoring enable decentralization and virtualization [17]. The LPS method such as autonomation is an example of a real-time capability that can identify defects on any scale of production. The additional effort involved in quality checking is unnecessary if it is built-in to the production line.

The methods of creating standards could help to achieve departmental transparency by the unambiguous procedures between stakeholders in a value chain. Standards are created by the agreement of stakeholders which could also increase transparency.

Thus, transparency could be enabled with a foundation of Lean principles, such as pull and flow production, zero defect, and standardization.

3.3. Lean Decentralization

A design requirement for a flexible I4.0 requires decentralized decision-makers to support cyber-physical systems and self-organizing production environments [20]. Decentralized decision-making can conflict with an organization's best interests due to the lack of shared goals and values. However, decisions made on limited pertinent information can be detrimental to an organization, whether it be a human or automated process [19]. The horizontal manufacturing process under I4.0 enables better decision-making and dynamic process control that provides the flexibility to execute last minute changes, or to alter the process after the customer decoupling point [17]. LPS shows that the beliefs of the company should be at the forefront of decision making. If all decision makers within an enterprise share the values of the organization, there is a shift to a bottom-up perspective which supports decentralized problem solving [39]. The employee and customer orientation of LPS fosters trust and learning opportunities to create responsible decentralized teams that can be the foundation for new standards.

Other than human based decisions, machine decisions and interactions built into a process could help to achieve decentralized working environment. Autonomation of the zero defect principle is one such method which makes decisions based on quality checks.

Thus, the decentralization design principle could be enhanced through Lean principles like employee orientation, and zero defect.

3.4. Lean Stakeholder orientation

A stakeholder orientation helps to understand groups of people who are affected by products or services and provides a basis for understanding how value is created [40]. The fulfilment of only what is expected from stakeholders leads to new service-based business models, where customers only pay for results. The interactions that occur between stakeholders can be optimized in VSM by process visualization and elimination of waste sources [40]. Similarly, the enterprise value stream map identifies sources of waste and opportunities for value creation on a vertical enterprise level. EVSM is defined at a higher level and is more general than a product value stream. It emphasizes enterprise-wide processes by addressing multiple programs or lines of business, rather than individual functions, programs, or tasks. It delivers value to all stakeholders and not just the end customer. Identification of enterprise waste with EVSM will assist in the vertical integration to improve efficiency as well as the ability to deliver value to all stakeholders [41]. This approach will try to overcome barriers faced while integrating horizontally and vertically, as all stakeholders will be considered to reach a common future stated goal, rather than an individual or product-based goal.

The stakeholder orientation is not just focused horizontally and vertically, but also internally towards the employees of an organization. The employees are the driving force of continuous improvement that help to improve quality and productivity in an organization.

Thus, optimization through Pull and flow production principle and culture change through employee orientation under Lean foundation assist to achieve stakeholder orientation of I4.0.

4. Conclusion

Companies strive to achieve a competitive edge and stay ahead in an era where change is happening quickly. This paper presented the shortcomings of I4.0 and built a design model to improve the implementation of I4.0 by applying Lean principles. The design model showed how LPS can be used as a foundation to achieve flexible and less complex systems of I4.0. The pillars of interoperability, transparency, decentralization, and stakeholder orientation provide a structure for a Lean integrated I4.0 system. The authors' perspective is that in the construction of new businesses and business models, LPS should not be overlooked in the creation of smart factories. The design model is still yet to be verified through application-oriented research.

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